

Stage 1: Logical on $\mathcal{C}$

L1: Adjunction
existence

L2: Dagger
involution

*15 ops total ($6 L + 4 I + 5 A$)
4 stages, linear ordering forced
L3-L6 require A1 in place
L4 requires L3 (cart. closure)*

Stage 2: Inductive

I1: Colimit
cocompletion

I2: Iteration
initial algebra

I3: Classifying
nerve/realization

I4: Yoneda
presheaf embed

Stage 3: Algebraic enrichment

A1: Monoidal
 $\otimes$, unit I

A2: Monoidal
 T, η, μ

A3: Dagger-comonads,
duals, snake

A4: Enrichment
hom-object

A5: Arithmetic
Gödel coding

Stage 4: Logical on enriched $\bar{\mathcal{C}}$

L3: Cartesian comonoids
exponentials

L6: Paraconsistency
negation

L4: Lawvere
self-modeling

L5: Frobenius
 $\mu \circ \delta = \text{id}$

12 Primitives $\rightarrow$ Crystal of 17,280,000 Types